

Project Submission Summary Report

Project Name: Power Query Data Transformation Project

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1. Sources Used

Data Sources:

- Country GDP Dataset (web/text source)
- Sales Excel Files: sales_jan.xlsx, sales_feb.xlsx, sales_march.xlsx, sales_apr.xlsx
- Employee Dataset: employees_with_country.xlsx

Source Connection Method: Parameterized Folder Path using Power Query Parameters.

Description of Data:

The GDP dataset contains country-wise GDP values from IMF, World Bank, and United Nations sources. The employee dataset contains employee information such as Employee ID, Name, Department, Position, Joining Date, Salary, Reason, and Country. The sales datasets contain Order ID, Date, Sales Person, Product, Quantity, Unit Price, and Total Amount.

2. Transformations Applied (Power Query)

- Removed blank rows and blank columns.
- Promoted first row as headers.
- Renamed columns with meaningful names.
- Changed data types for dates, salary, and sales values.
- Removed duplicate records and filtered null values.
- Rounded sales values to 2 decimal places.
- Created Cost and Profit columns.
- Extracted Day, Month, Year, and Quarter from Date.
- Created Fiscal Month column.
- Calculated Years of Experience using Joining Date.
- Created Sales Category using Conditional Columns (High, Medium, Low).
- Added Index Columns (0-based and 1-based).
- Performed Pivoting and Unpivoting operations.
- Appended January–April sales files using Append Queries as New.
- Merged Employee and Sales data using Sales Person and Employee Name.
- Grouped sales data by Country and calculated Total Sales, Average Order Value, and Transaction Count.
- Enabled Column Quality, Column Distribution, and Column Profile.
- Created FolderPath parameter for dynamic source loading.
- Configured Data Source Settings and credentials.
- Performed Refresh Simulation by adding sales_apr.xlsx.

3. Challenges Faced & How They Were Solved

Challenge 1: Employee Name Matching During Merge

Issue: Sales data contained first names while employee data stored employee details separately.

Solution: Created a helper Employee_Name column and used Merge Queries with fuzzy matching.

Challenge 2: Missing Country Column After Merge

Issue: Country information was not visible after merging.

Solution: Expanded the merged Employees column and selected Country, Department, Position, and Employee ID fields.

Challenge 3: Group By Configuration

Issue: Multiple columns were mistakenly selected during Group By operation.

Solution: Grouped only by Employees.Country and applied aggregation functions correctly.

Challenge 4: Dynamic Data Refresh

Issue: New monthly files required manual imports.

Solution: Implemented FolderPath parameter and folder-based connection for automatic refresh.

4. Final Deliverables File Check

- Power Query Project.pbix – Complete Power BI file containing all Power Query transformations.
- Power_Query_Detailed_Report.pdf – Project summary documentation.
- Sales datasets and employee dataset used for analysis and reporting.